DO NOW — RECALL

Recursion and algorithm tracing

Name: _____ Class: _____ Date: _____

1. What is an activation record?

2. When does code after recursive call execute?

3. What is auxiliary stack space for depth n?

4. Search for 12 in [2,4,6,8,10,12,14]. Midpoint indexes?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Saved data for an active function call.

2. Model answer

On return from the deeper call.

3. Model answer

O(n) call stack.

4. Model answer

3→5; index5 found.

By the end, you can…

• Trace recursive calls downward before calculating values on return.

• Name the base case and variable that progresses toward it.

• Recursion depth consumes call-stack memory even when total time is linear.

Vocabulary — know these terms

recursion — Subroutine calls itself.

base case — Stops additional calls.

call stack — Active subroutine frames.

unwinding — Returning from nested calls.

NEW LEARNING

Read, annotate and explain

Understand

Trace recursive calls downward before calculating values on return.

Apply

Name the base case and variable that progresses toward it.

Evaluate

Recursion depth consumes call-stack memory even when total time is linear.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Predict output of printing before a recursive call for n=3.

Explain the result

Outputs 321 on descent. Nothing prints for n0.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace SumOdds(4), listing contributions and returned result. [6]

Question 2

For F(n): if n=0 return, else print(n) then F(n-1), state F(4) output and why. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Write an iterative C# method to sum the positive odd integers up to n (n ≥ 0), matching SumOdds(n). Compare its time and auxiliary space with the recursive method. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Printing after the recursive call appears on stack unwinding.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.